

Rishika Mamidibathula

rm4318@columbia.edu · (332) 251-6686 · [linkedin.com/in/rishika-mamidibathula](https://www.linkedin.com/in/rishika-mamidibathula) · rishika-m.com · github.com/rishika1099

EDUCATION

Columbia University

New York, NY

M.S. Data Science, GPA: 3.9/4.0

Aug 2025 – Expected Dec 2026

- Coursework: Deep Learning, Generative AI Systems, Agentic AI, High Performance Computing, Causal Inference, Statistical Inference
- Teaching Assistant for Natural Language Processing and AI for Public Policy; Data Science Institute Student Council Member

Vellore Institute of Technology

Vellore, IN

B.Tech. Computer Science and Engineering (Data Science specialisation), GPA: 4.0/4.0

Jul 2019 – May 2023

- Ranked 7th out of 200 (Top 4%); Merit Scholarship recipient (2019–2023); Program Representative (2019–2023)
- Coursework: Machine Learning, Artificial Intelligence, Natural Language Processing, Image Processing, Social Information Networks

SKILLS

Programming & Frameworks: Python, SQL, R, C++, PyTorch, TensorFlow, scikit-learn, FastAPI, NumPy, Pandas

LLM & AI Systems: RAG, LangChain, Hugging Face, ChromaDB, FAISS, Multimodal AI, Prompt Engineering, Agent Workflows

Data Engineering & Infrastructure: Databricks, PySpark, MongoDB, MySQL, BigQuery, Redis, Docker, Git, REST APIs

Cloud & MLOps: AWS, Azure DevOps, Weights & Biases, CI/CD, Experiment Tracking, Feature Engineering, Model Monitoring

WORK EXPERIENCE

NYC Administration for Children’s Services (ACS)

New York, NY

Predictive Analytics Intern

Jun 2026 – Aug 2026

- Built an explainability framework for a child welfare risk model on 900K+ records, transforming black-box predictions into transparent, policy-driven decisions through configurable operating thresholds spanning 25–80% recall (0.799 ROC-AUC, 7.2× base-rate precision).
- Designed a three-level reliability audit proving explanations reproducible across evaluation sets ($\rho=0.991$), then established via noise-control testing that only 15 of 149 features carried real signal, setting a defensible limit on what the agency can claim from any feature ranking.

Shell

Bengaluru, IN

Software Engineer

Aug 2023 – Jul 2025

- Developed machine learning forecasting solutions in Databricks (PySpark, SQL) across 12 business units, reducing forecast error by 23% and enabling \$100K+ annual cost optimization decisions.
- Built production data pipelines, feature engineering workflows, and 5 Blue Prism RPA bots with logging and retry logic, cutting manual reporting effort by 85% (120+ hours/quarter) and improving SLA compliance from 92% to 99%.

Novartis

Hyderabad, IN

Technical Analyst Intern

Jan 2023 – Jul 2023

- Engineered a NLP-driven clinical trial analysis workflow for drug-level sentiment mining, automated summarization, and outcome extraction from unstructured research documents, reducing manual literature review and evidence synthesis effort by 40% per quarter.
- Built predictive and time-series forecasting pipelines on environmental operations data, integrating web-scraped external signals, feature engineering, and temporal trend analysis to support sustainability initiatives targeting a 19% annual reduction in carbon emissions.

RESEARCH EXPERIENCE AT COLUMBIA UNIVERSITY

Lamont-Doherty Earth Observatory, McKinley Group

Palisades, NY

Data Scientist – Ocean Carbon Data Valuation

Sep 2026 – Present

- Extending a Data Shapley framework valuing individual SOCAT surface ocean fCO₂ observations against a physical model prior rather than an empty baseline, using KNN-Shapley approximation and spatially blocked cross-validation to target where sparse, expensive measurements would most improve the ocean carbon sink reconstructions contributed annually to the Global Carbon Budget.

Irving Medical Center, AI4VS Lab

New York, NY

Computer Vision Engineer – Ophthalmic AI & Eye Tracking

Jul 2026 – Present

- Developing gaze-augmented deep learning models for glaucoma diagnosis that fuse clinician eye-tracking with OCT report features, aligning model attention with expert reading behavior to make automated screening for irreversible vision loss auditable enough for clinical use.

Irving Medical Center

New York, NY

AI Engineer – Clinical LLM & Phenotyping

Jan 2026 – Present

- Built an extraction system with hybrid de-identification, longitudinal EHR reconstruction, and hallucination-aware validation, converting unstructured notes into 56 cardiac sarcoidosis phenotype variables and replacing months of chart review with a manuscript-ready cohort.

Graduate School of Arts and Sciences

New York, NY

Machine Learning Engineer – Human Rights LLM Evaluation

Jan 2026 – Present

- Developed a two-stage retrieval-augmented LLM framework with automated web search and chain-of-thought reasoning scoring 27 defense manufacturers across 9 human-rights dimensions, cutting analyst review effort by 80% with kappa and Krippendorff’s alpha validation.

PROJECTS

Folio: Clinical Multimodal RAG

folio-health.vercel.app

- Built a multimodal clinical platform unifying RAG, document understanding, speech transcription, and longitudinal record management across text, PDF, image, and audio, hitting 85.1% extraction micro-F1 and 100% RAG recall@1 at sub-2s latency.

Federal Eagle: Multi-Agent Legal Assistant

federal-eagle.streamlit.app

- Orchestrated a multi-agent CrewAI system for U.S. federal legal analysis with LangChain and ChromaDB retrieval over all 54 U.S. Code titles (200K+ documents), precedent search, elements analysis, and draft generation, reaching 87% Precision@5 on benchmark legal queries.

KV Cache Optimization for LLM Inference

github.com/rishika1099/KV-Cache-Optimization

- Benchmarked KV-cache optimizations for Llama-2-7B including KIVI quantization, TopK sparse selection, SnapKV eviction, and MLA latent compression with custom Triton kernels, achieving 4× compression at LongBench quality parity and a 3.1× peak throughput gain.